

Difficulty Analysis of Posiform-Planted QUBO Benchmarks via Energy and Hamming Metrics

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Abstract—Benchmarking quantum annealers and classical optimization heuristics requires problem instances whose optimal solution is known in advance, yet that remain genuinely hard to solve. Posiform planting is a recent technique that generates Quadratic Unconstrained Binary Optimization (QUBO) instances with a single, unique planted optimum; a hardened variant raises their difficulty by adding random spin-glass terms and scaling the planted component by a coefficient α . A practical obstacle, however, is measurement: hardness is conventionally graded by the ground-state success rate, but across the entire hard regime this rate is identically 0%, so it cannot distinguish instances, solvers, or values of α from one another. This poster presents a measurement protocol that restores discrimination in the zero-success regime. Instead of asking whether a run succeeded, we quantify how close the failing runs come to the planted optimum, using two complementary quantities computed over failures only and on a fixed set of the hardest instances: the energy gap to the optimum and the Hamming distance from the planted answer. Together they convert the flat 0% rate into a continuous, rankable difficulty curve. On instances defined over a D-Wave Pegasus graph ($n = 5612$ qubits), the curve is non-monotone in α – rising to a distinct peak and then declining – and the same structure reproduces on a D-Wave Advantage quantum annealer, indicating that it reflects the benchmark instances rather than any single solver. The protocol offers a finer-grained way to grade planted QUBO benchmarks.

Index Terms—QUBO benchmarking, quantum annealing, simulated annealing, posiform planting, performance evaluation, NISQ, quantum optimization.

uninformative, yet still rank instances, solvers, and a hardware-relevant tunable difficulty parameter. The study spans both classical (simulated annealing) and quantum (D-Wave Advantage) solvers under a common measure, supporting cross-platform comparison. It thus connects to QCE topics in (i) quantum benchmarking and performance characterization, (ii) quantum annealing and optimization, and (iii) reproducible evaluation methodology and tooling for quantum hardware. All code and benchmark instances are released openly (github.com/YIDEUNKIM/qubo_dataset) to support reproducibility.

POSTER RELEVANCE TO IEEE QUANTUM WEEK (QCE
2026)

This work addresses the *benchmarking, performance evaluation, and verification of programmable quantum annealers* – core themes of IEEE Quantum Week. Its subject is the QUBO problem, the native input format of D-Wave quantum annealers and a central model for quantum and quantum-inspired optimization. The proposed failure-conditioned metrics target the NISQ-era regime in which solvers seldom reach the true ground state and a binary success rate is

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